

Problemas Tema 4 Gonzalez Valentin Adrian

1.- Resolver la siguiente integral por los métodos Solicitados

$$\int_0^3 (x^2 e^x) dx = 98.42768$$

Regla del Trapecio.

$$I = (b-a) \left(\frac{f(x_0) + f(x_1)}{2} \right)$$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(3) = 180.76983$$

$$I = (3-0) \left(\frac{0 + 180.76983}{2} \right) = 271.15474$$

$$Er\% = \frac{198.42768 - 271.15474}{198.42768} \times 100 = 175\%$$

Trapecio multisegmento. $n=4$

$$I = (b-a) \left(\frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n} \right)$$

$$h = \frac{3-0}{4} = 0.75$$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(0.75) = 1.19081$$

$$f(x_2) = f(1.5) = 10.08380$$

$$f(x_3) = f(2.25) = 48.03166$$

$$f(x_n) = f(3) = 180.42768$$

$$I = \frac{3}{2(4)} (0 + (1.19081 + 10.08380 + 48.03166) + 180.42768) \\ = 89.90023$$

$$Er \% = \frac{98.42768 - 89.90023}{98.42768} \times 100 = 8.6 \%$$

Simpson 1/3.

$$I = (b-a) \left(\frac{f(x_0) + 4f(x_1) + f(x_2)}{6} \right)$$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(1.5) = 10.08380$$

$$f(x_2) = f(3) = 180.42768$$

$$I = 3 \left(\frac{0 + 4(10.08380) + 180.42768}{6} \right) = 110.38144$$

$$Er \% = \frac{98.42768 - 110.38144}{98.42768} \times 100 = 0.12 \%$$

Simpson 1/3 multisegmento. $n=4$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(0.75) = 1.19081$$

$$f(x_2) = f(1.5) = 10.08380$$

$$f(x_3) = f(2.25) = 48.03166$$

$$f(x_4) = f(3) = 180.38144$$

$$I = (b-a) \left(f(x_0) + 4 \sum_{j=1,3,5} + 2 \sum_{j=2,4,6} f(x_j) + f(x_n) \right)$$

$$I = 3 \left(0 + 4(1.19081 + 48.03166) + 2(10.09380) + 180.42768 \right) = 99.37129$$

$$Er\% = \frac{98.42768 - 99.37129}{98.42768} \times 100 = 0.9\%$$

Simpson 3/8.

$$I = (b-a) \left(f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) \right)$$

$$h = \frac{(3-0)}{3} = 1$$

$$f(x_0) = f(0) = 0$$

$$f(x_1) = f(1) = 2.71828$$

$$f(x_2) = f(2) = 29.55622$$

$$f(x_3) = f(3) = 180.76983$$

$$I = 3 \left(0 + 3(2.71828) + 3(29.55622) + 180.76983 \right) = 104.09692$$

$$Er\% = \frac{98.42768 - 104.09692}{98.42768} \times 100 = 5.7\%$$